Madhya Pradesh – Grade 10
2015 – Mathematics

Choose the correct option and write it in your answer-book: **$1 \times 5 = 5$**

Question: If $a_1b_2 \neq a_2b_1$ then the system of equations $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$:

Options:

a. Has a unique solution

b. Has no solution

c. Has infinitely many solutions

d. Has two solutions.

Answer: (a)

Solution: $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ condition hold for unique solution.

Question: The coordinates of intersection points of lines x + y = 7 and x – y = 3 will be :

Options:

a. (4, 3)

b. (7, 4)

c. (5,2)

d. (6, 1)

Answer: (c)

Solution:

$x + y = 7$

$\underline{x - y = 3}$

$2x = 10$

$x = 5$

y = x – 3

= 5 – 3 = 2

Question: Zeros of $x^2 - 2x$ are:

a. -2,

b. 2, -2

c. 0,2

d. 1,2

Answer: (c)

Solution:

$x^2 - 2x$

x(x – 2)

x = 0   x – 2 = 0

$x = 2$

Question: If P = $\frac{1}{x+1}$ and Q = $\frac{x^2-1}{x-1}$, then value of P.Q is:

a. x + 1
b. 1
c. x – 1
d. x^2-1

Answer: (b)

Solution:

P. Q

$$= \frac{1}{x+1} \times \frac{x^2-1}{x-1}$$

$$= \frac{1}{x+1} \times \frac{(x+1)(x-1)}{x-1} = 1$$

Question: The third proportional to 8, 12 is:

a. 18
b. 8
c. 4
d. 20

Answer: (a)

Solution: Third proportional says

a : b : : b : c

8 : 12 = 12 : c

$$\frac{8}{12} = \frac{12}{c}$$

$$c = \frac{12 \times 12}{8}$$

c = 18

Question:

Match the Followings:

Column 'A'	**Column 'B'**
1. $\sin 30°$	(a) 0
2. $\sqrt{\sec^2\theta - 1}$	(b) $\operatorname{cosec}\theta$
3. $\sin 55° - \cos 35°$	(c) $\sin\theta$
4. $\frac{\sec\theta}{\tan\theta}$	(d) $\cot\theta$
5. $\frac{\sqrt{1-\sin^2\theta}}{\sin\theta}$	(e) $\cos 60°$
	(f) $\tan\theta$

Answer:
1 – e
2 – f
3 – a
4 – b
5 – d

Solution:
1.
$\sin 30 = \frac{1}{2}$ *also* $\cos 60 = \frac{1}{2}$
Both are equal
2.
$\sec^2\theta - \tan^2\theta = 1$
$\sec^2\theta - 1 = \tan^2\theta$
$\tan\theta = \sqrt{\sec^2\theta - 1}$
3. sin55 – sin(90 – 35)
= sin55 – sin55
= 0
4.
$\frac{1}{\cos\theta} \times \frac{\cos\theta}{\sin\theta}$
$= \frac{1}{\sin\theta} = \text{cosec}\theta$
5.
$\sqrt{\frac{1-\sin^2\theta}{\sin\theta}} = \frac{\cos\theta}{\sin\theta}$
$= \cot\theta$

Question:
Fill in the blanks:
1.An equation whose maximum degree of variable is two is called ………. Equation.
Answer: quadratic
Solution: The highest degree of 2 represents quadratic

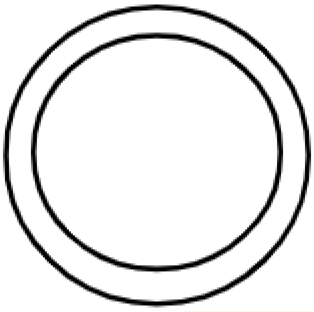
2. The reduction in price of the article with time is called …….. .
Answer: Depreciation

3. If the corresponding angles of two triangles are equal then the triangles are ……..
Answer: Similar
Solution: Because corresponding angles in triangles are equal it represents similarity

4. The solid bounded by two concentric sphere is called …………

Answer: Spherical shell

Solution:



← The solid is termed as spherical sheet

Question: The line segment joining the two point on the circumference of the circle is called ……

Solution: Chord

Write True/False in the following: **1 × 5 = 5**

1. The service tax is an indirect tax

Answer: True

2. The statement of Thales theorem is: "If a line divides any two sides of a triangle in the same ratio, then the line must be parallel to the third side."

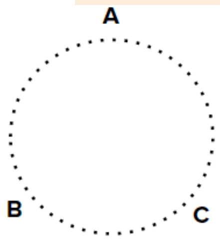
Answer: False

3. A circle can be drawn passing through three non collinear points.

Answer: True

Solution:

For example



4. The length of two tangents drawn from an external point to a circle are unequal.

Answer: False

Solution: Length of two tangents drawn from an external point to a circle are always equal.

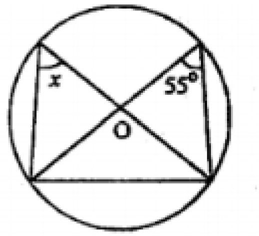
5. A line joining the object under consideration with eye is known as line of sight.

Answer: True

Write the answer is one word/sentence each:

1. What is the present rate of education cess?
Solution: 3%

2. What will be the value of x in the figure given bellow:



Answer: $55°$
Solution: Angle subtended in circle from same chord are always equal (property of chord)

Question: Write the ratio between the volumes of cylinder and cone which have same radius and height.
Answer: 3 : 1
Solution: Volume of cylinder = $\pi r^2 H$

$$\text{Volume of cone} = \frac{1}{3}\pi r^2 H$$

$$\frac{\text{Volume of cylinder}}{\text{Volume of cone}} = \frac{\pi r^2 H}{\frac{1}{3}\pi r^2 H} = 3:1$$

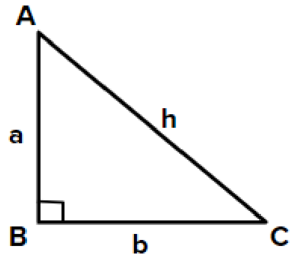
Question: Write the probability of sure event
Answer: 1

Question: Find the mode of the following observation: 2, 3, 4, 2, 12, 8, 7, 9, 8, 6, 8, 5, 8.
Answer:8
Solution: Mode represent the number having highest frequency

Question: (a) Write the statement of Pythagoras theorem

Or

(b) Check whether 8 cm, 15 cm and 17 cm are the sides of right angled triangle.
Solution:
(a) In a right angled triangle the addition of square of perpendicular sides equals to the hypotenuse square
So, $a^2 + b^2 = h^2$



(b) On applying Pythagoras on 8, 15, 17

$$8^2 + 15^2 = 17^2$$

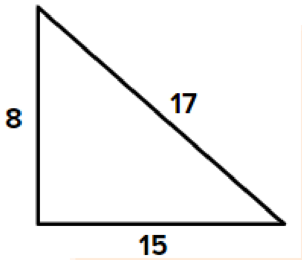
L.H.S $\quad 8^2 + 15^2 = 64 + 225$

$$= 289$$

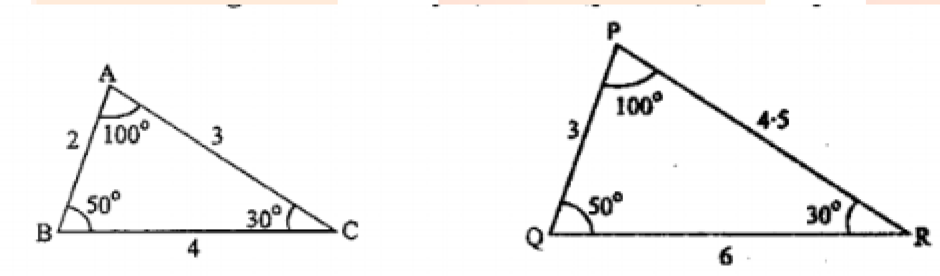
R.H.S $\quad 17^2 = 289$

$\because$ L.H.S $=$ R.H.S

Hence, 8, 15, 17 represents side of right angle triangle.

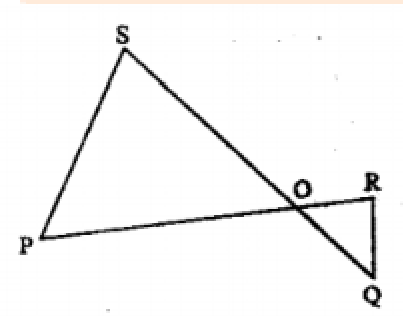


Question: Are these triangle similar? If yes, then why? If no, then why?



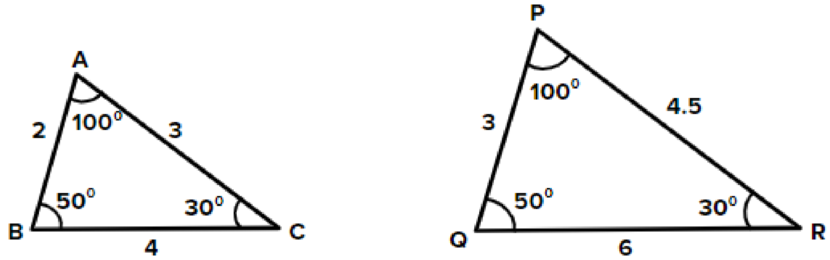
Or

In the figure given below $\Delta POS \sim \Delta ROQ$, Prove, that PS || QR.



Solution:

(a)



To Prove: $\Delta ABC \sim \Delta PQR$

Proof : $\angle A = \angle P$

$\angle B = \angle Q$

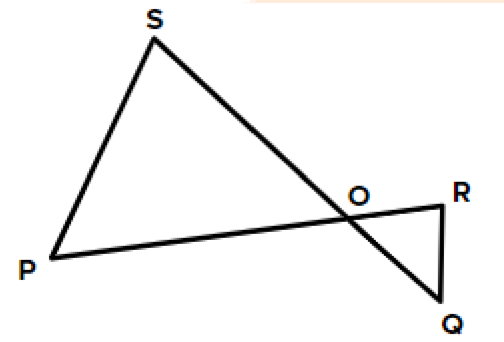
$\angle C = \angle R$

Also, $\frac{AB}{PQ} = \frac{AC}{PR} = \frac{BC}{QR}$

$\frac{2}{3} = \frac{2}{4.5} = \frac{4}{6}$

Hence, By AAA $\Delta ABC \sim \Delta PQR$

(b)



$\because \Delta POS \sim \Delta ROQ$

Hence the corresponding sides and corresponding angles will be equal

$\angle SOP = \angle ROQ$

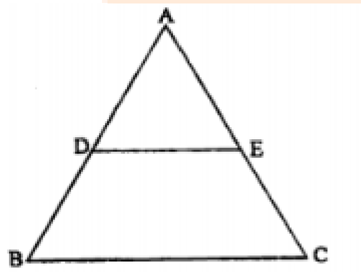
$\angle PSQ = \angle RQO$

$\angle SPR = \angle PRQ$

Alternately**,** SQ and PR acts as transversal and $\angle$PSQ and RQS, $\angle$SPR and PRQ acts as alternate angles.

Hence, PS || QR

Question: In the figure given below DE || BC. If $\frac{AD}{DB} = \frac{2}{5}$ and EC = 10 cm find AE.



or

If the areas of two similar triangles are equal. Prove that the triangle are congruent.

Solution:

(a)

In ΔDE || BC

So, using corresponding ratio

$$\frac{AD}{DB}=\frac{AE}{EC}$$

$$\frac{2}{5}=\frac{AE}{10}$$

$$AE=\frac{20}{5}=4\,cm$$

(b)

Given:

$\Delta ABC \sim \Delta PQR$ and, Area $\Delta ABC =$ Area ΔPQR

To Prove: $\Delta ABC \cong \Delta PQR$

Proof: $\frac{Ar\,\Delta\,ABC}{Ar\Delta\,PQR}=1$

$$\frac{AB^2}{PQ^2}=\frac{BC^2}{QR^2}=\frac{CA^2}{PR^2}=1$$

Using theorem of area of similar triangle

$\Rightarrow$ AB = PQ, BC = QR, CA = PR

Thus, $\Delta ABC \cong \Delta PQR$ by SSS criteria.

Question: Find the median of the following values of variate: 15, 35, 18, 26, 19, 25, 29, 20, 27.

Or

Find the mean of all factors of 20.

Solution:

(a) Arrange data in ascending order

15, 18, 19, 20, 25, 26, 27, 29, 35

No. of observation = 9 (odd)

Median = $\left(\frac{n+1}{2}\right)$th observation

$= \left(\frac{9+1}{2}\right)$th 5^{th} observation

Hence, 25 is median

(b)

Mean = $\frac{\text{Total of numbers}}{\text{Number of observation}}$

Factors of 20 = 1, 2, 4, 5, 10, 20

Total of numbers = 1 + 2 + 4 + 5 + 10 + 20

= 42

Number of Observation = 6

Mean = $\frac{42}{6}$ = 7

Question: If the probability of raining tomorrow is $\frac{2}{3}$ then what will be the probability of not raining tomorrow.

Or

Two coins are tossed simultaneously. Find the probability of getting Head on one coin and Tail on another coin.

Solution:

(a)

Probability of raining tomorrow = $\frac{2}{3}$

Total probability = 1

Probability of not = 1- $\frac{2}{3}$ = $\frac{1}{3}$

(b)

Two coin sample space = {HH, HT, TH, TT}

Events = {TH, HT}

$$\text{Probability} = \frac{\text{Event}}{\text{sample space}}$$

$$= \frac{2}{4} = \frac{1}{2}$$

Question: Solve the following system of equation by elimination method:

3x + 2y = 11, 2x + 3y = 4.

Or

Solve the following system of equations by Paravartya method of Vedic mathematics:

2x + y = 5, 3x – 4y = 2

Solution:

(a)

$3x+2y=11\times 3........(1)$

$2x+3y=4\times 2.........(2)$

$$\begin{array}{l} 9x+6y=33 \\ 4x+6y=8 \\ \underline{-\quad -\quad\ -} \\ \ \ 5x\ = 25 \\ \quad x=5 \end{array}$$

$$y=\frac{11-3\times 5}{2}$$

$= \frac{-4}{2}$

y = – 2

(b)

$2x + y = 5 \times 4$........(1)

$3x - 4y = 2$...........(2)

$$\begin{array}{l} 8x + 4y = 20 \\ \underline{3x - 4y = 2} \\ 11x \ = 22 \\ \quad x = 2 \end{array}$$

y = 5 – 2x

= 5 – 2 × 2

= 1

Question: The sum of two numbers is 80 and the first number is 20 more than the second. Find the numbers

or

The cost of 2 chairs and 3 tables is Rs. 800 and the cost of 4 chairs and 3 tables is Rs. 1000. Find the cost of 2 chairs and 2 tables.

Solution:

(a)

Let number be x and y

x + y = 80……. (1)

x = 20 + y …… (2)

$$\begin{array}{l} x - y = 20 \\ \underline{x + y = 80} \\ 2x = 100 \\ \quad x = 50 \end{array}$$

y = x – 20

= 50 – 20

y = 30

(b)

Let cost of chair be x and table be y

$$\begin{array}{l} 2x + 3y = 800 \text{ ....(1)} \\ 4x + 3y = 1000 \text{.........(2)} \\ \underline{- \quad - \quad\quad -} \\ -2x = -200 \\ \quad x = 100 \end{array}$$

y = $\frac{800 - 2x}{3}$ from equation 1

$= \frac{800-200}{3}$

y= 200

cost of 2 chair + 2 TABLE

$= 2\times100+2\times200$

= 600 Rs

Question: If $\frac{x}{a}=\frac{y}{b}=\frac{z}{c}$ then prove that $\frac{x^3}{a^3}-\frac{y^3}{b^3}+\frac{z^3}{c^3}=\frac{xyz}{abc}$

Or

If q is the mean proportional of p and r then prove that:

$$p^2-q^2+r^2=q^4\left(\frac{1}{p^2}-\frac{1}{q^2}+\frac{1}{r^2}\right)$$

Solution:

(a)

$\frac{x}{a}=\frac{y}{b}=\frac{z}{c}=k$ (let)

x = ak, y = bk, z = ck

L.H.S: $\frac{x^3}{a^3}-\frac{y^3}{b^3}+\frac{z^3}{c^3}$

$=\frac{a^3k^3}{a^3}-\frac{b^3k^3}{b^3}+\frac{c^3k^3}{c^3}$

$=k^3-k^3+k^3=k^3$

R.H.S: $=\frac{\text{xyz}}{\text{abc}}=\frac{\text{(ak) (bk) (ck)}}{\text{abc}}$

$=k^3$

(b)

p : q :: q : r

$\frac{p}{q}=\frac{q}{r}$ $q^2=pr$

R.H.S: $q^4\left(\frac{1}{p^2}-\frac{1}{q^2}+\frac{1}{r^2}\right)$

$=q^4\left(\frac{1}{p^2}-\frac{1}{pr}+\frac{1}{r^2}\right)$

$=q^4\left(\frac{r^2-pr+p^2}{p^2r^2}\right)$

$=q^4\left(\frac{r^2-pr+p^2}{q^4}\right)$

$=p^2-pr+r^2$ = L.H.S

$=p^2-q^2+r^2$ proved

Question: Solve the equation $x^2 - 5x - 6 = 0$ by formula method.

or

Find the value of p in the equation $2py^2 - 8y + p = 0$ so that the equation has equal roots.

Solution:

(a) Formula method $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$x^2 - 5x - 6 = 0$$

a= 1 b = -5 c = -6

$$x = \frac{5 \pm \sqrt{(-5)^2 - 4 \times 1 \times -6}}{2 \times 1}$$

$$= \frac{5 \pm \sqrt{25 + 24}}{2 \times 1}$$

$$= \frac{5 \pm 7}{2}$$

= 6, -1

(b) For equal roots $\Delta = b^2 - 4ac = 0$

$$2py^2\text{-}8y\text{+}p = 0$$

a = 2p b = -8 c = p

$$0 = (-8)^2 - 4 \times 2p \times p$$

$$0 = 64 - 8p^2$$

$$= 8(8 - p^2)$$

$$p^2 = 8$$

$$p = \pm 2\sqrt{2}$$

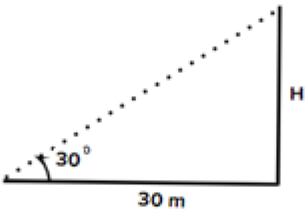
Question: At a distance of 30 m away from the tower, the angle of elevation of the top of the tower is $30°$. Find the height of the tower.

Or

From the top of 60m high house, the angle of depression of the ship is $60°$. Find the distance between the ship and the foot of the light house.

Solution:

(a)

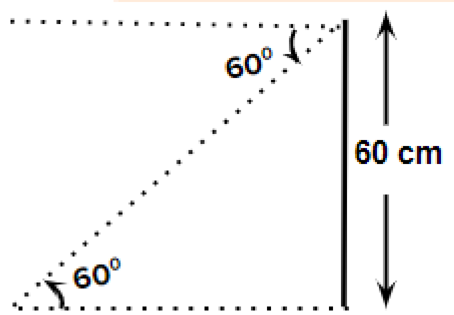


$$\tan 30 = \frac{H}{30}$$

$$H = \frac{1}{\sqrt{3}} \times 30$$

$$= 10\sqrt{3}\ \text{m}$$

(b)



$$\tan 60 = \frac{60}{x}$$

$$x = \frac{60}{\sqrt{3}}$$

$$= \frac{3 \times 2 \times 10}{\sqrt{3}}$$

$$= 20\sqrt{3}\ \text{m}$$

Question: In a circle an arc subtends an angle $45°$ at the center. If the length of arc is 11 cm then find the radius of circle.

Or

If V is the volume of cuboid whose length is 'a', breadth is 'b' and height is 'c' and 's' is its surface area then prove that: $\frac{1}{V} = \frac{2}{S}\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$

Solution:

(a) Length of arc = $\frac{\theta}{360} \times 2\pi r$

Length of arc = $\frac{45}{360} \times 2 \times \frac{22}{7} \times r$

$$r = 7 \times \frac{360 \times 11}{45 \times 22 \times 2}$$

$r = 14$ cm

(b)

Length = a Breadth = b height = c

Volume of cuboid = $a \times b \times c = abc = v$

Surface area (S) = (lb + bh + hl) $\times$ 2

= (ab + bc + ca) $\times 2$

R.H.S = $\frac{2}{S}\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$

$= \frac{2}{2\times(ab+bc+ca)}\left(\frac{bc+ac+ab}{abc}\right)$

$= \frac{2}{2abc}$

$= \frac{2}{2v} = \frac{1}{v}$ = L.H.S

Question: The radius of a cone is 7 cm and its height is 9 cm. The volume of this cone is equal to lateral surface area of another cone which has same radius. Find the slant height of the cone.

Or

A cylinder of height 90 cm and base diameter 8 cm is melted and recast into spheres of diameter 12 cm. Find the number of spheres.

Solution:

(a) Volume of cone = $\frac{1}{3}\pi r^2 h$

Radius = 7 cm Height = 9 cm

$$v = \frac{1}{3}\times\frac{22}{7}\times 7\times 7\times 9$$

$= 22\times 21 \text{cm}^3$

From other cone

Radius = 7 cm slant height = l

Surface Area = πrl

Now, $22\times 21 = \pi r l$

$$\frac{7\times22\times21}{22\times7}=l$$

l = 21 cm

(b) Cylinder: Radius = $\frac{d}{2}=\frac{8}{2}=4\text{cm}$

Height = 90 cm

Volume of cylinder = $\pi r^2 h$

$$=\frac{22}{7}\times4\times4\times90$$

$$= 1440\pi \text{ cm}^3$$

Sphere: diameter = 12 cm

Volume of sphere = $\frac{4}{3}\pi r^3$

$$=\frac{4}{3}\times\pi\times6^3$$

$$=\frac{4}{3}\times6\times6\times6\times\pi$$

$$= 48\times6\pi \text{ cm}^3$$

No. of sphere = $\frac{\text{Total volume of melt}}{\text{Volume of 1sphere}}$

$$=\frac{1440\pi}{48\times6\pi}=5$$

Question: Find cyclic factors: $x(y^2+z^2)+y(z^2+x^2)+z(x^2+y^2)+2xyz$

Or

Which rational expression should be subtracted from $\frac{x^2+1}{x-1}$ get $\frac{x-3}{x+1}$?

Solution:

(a) $x(y^2+z^2)+y(z^2+x^2)+z(x^2+y^2)+2xyz$

By factor theorem, x + y is a factor of the given expression because

$$(-y)\left(y^2+z^2\right)+y\left(z^2+y^2\right)+z\left(y^2+y^2\right)+z(-y)yz=0$$

Since x + y is a factor, y + z and z + x are also factors.

(b) let z rational is subtracted

$$\frac{x^2+1}{x-1}-z=\frac{x-3}{x+1}$$

$$\frac{x^2+1}{x-1}-\frac{x-3}{x+1}=z$$

$$\frac{x^3+x+x^2+1-(x^2-3x-x+3)}{x^2-1}=z$$

$$\frac{x^3+x+x^2+1-x^2+4x-3}{x^2-1}=z$$

$$\frac{x^3+5x-2}{x^2-1}=z$$

Question: If α and β are roots of quadratic equation $ax^2+bx+c=0$, then find $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}$.

Or

The length of rectangle is 5 cm more than its breadth. If the area of the rectangle is 150 sq. cm. Find the sides of rectangle.

Solution:

(a)

$$ax^2+bx+c=0$$

$$\alpha+\beta=\frac{-b}{a} \quad \alpha\beta=\frac{c}{a}$$

To find:

$$\frac{\alpha}{\beta}+\frac{\beta}{\alpha}$$

$$=\frac{\alpha^2+\beta^2}{\alpha\beta}$$

$$=\frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}$$

$$=\frac{\frac{b^2}{a^2}-\frac{2c}{a}}{\frac{c}{a}}=\frac{\frac{b^2-2ac}{a^2}}{\frac{c}{a}}=\frac{b^2-2ac}{a}\times\frac{1}{c}$$

$$=\frac{b^2-2ac}{ac}$$

(b) length = Breadth + 5

$l = b + 5$

Area = 150

$b \times l = 150$

$b^2+5b-150=0$

$b^2+15b-10b-150=0$

$b(b+15)-10(b+15) = 0$

$(b+15)(b-10) = 0$

b = 10 cm (satisfied)

length = 10 + 5 = 15 cm

Question: Find the compound interest on Rs. 8,000 for the period of $1\frac{1}{2}$ years at the rate of interest 10% per annum, if the interest is compounded half yearly.

Or

A watch is sold for Rs. 960 cash or for Rs. 480 cash down payment and two monthly instalments of Rs. 245 each. Find the rate of interest charged under the instalment plan.

Solution:

(a) Principal **= 8000**

Rate = 5% (half yearly)

Time = 3 (half yearly)

$$A = p\left(1+\frac{r}{100}\right)^n$$

$= 8000\left(1+\frac{5}{100}\right)^3$

$= 8000 \times \frac{21}{20} \times \frac{21}{20} \times \frac{21}{20}$

= 9261 Rs

CI = Amount – principal

= 9261 – 8000

= 1261 Rs

(b)

Price of watch = 960 Rs

Down payment = 480 Rs

Left out = 480 Rs

Total Instalment = 245 × 2 = 490

Total paid = 490 + 480

= 970 Rs

Interest paid = 970 – 960

= 10 Rs

I = $\frac{p \times r \times T}{100}$

10 = $\frac{490 \times r \times 1}{100 \times 12}$

r = 24.48 %

Question: Construct the incircle of the equilateral triangle whose one side is 8 cm and write the steps of construction also.

Or

Construct a cyclic quadrilateral ABCD in which vertical angle $\angle B = 65°$, AB = 4 cm. AC = 6 cm. AD = 4 cm. Write the steps of construction also.

Solution:

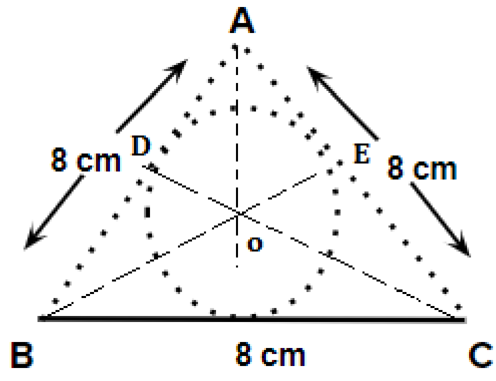
(a) Draw base = 8 cm

- Measure 8 cm using compass draw triangle

- Draw angle bisector of each angle

- Name the intersection as O

- Measure distance between O and any near side and draw the circle.



(b)

Steps:

- Draw AB = 4 cm using scale

- Draw $\angle B = 65°$ using compass

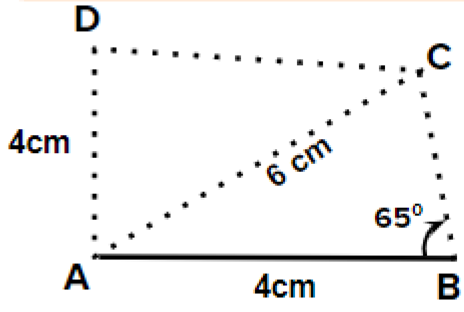
- Point at A measures 6 cm on compass draw an arc to get C.

- Extend line C.

- Measures 4 cm radius and draw an arc

- whatever this cut join D

- Join ABCD to get quadrilateral



Question: Prove that following identity: $\sqrt{\frac{1-\sin\theta}{1+\sin\theta}} = (\sec\theta - \tan\theta)$.

Or

Simplify: $(\sec\theta + \tan\theta)(1 - \sin\theta)$.

Solution:

(a) L.H.S $\sqrt{\frac{1-\sin\theta}{1+\sin\theta}\times\frac{1-\sin\theta}{1-\sin\theta}}$ (on Rationalising)

$$=\sqrt{\frac{(1-\sin\theta)^2}{1-\sin^2\theta}}=\frac{1-\sin\theta}{\cos\theta}=\frac{1}{\cos\theta}-\frac{\sin\theta}{\cos\theta}$$

$$=\sec\theta-\tan\theta \quad \text{(R.H.S)}$$

(b)

$(\sec\theta+\tan\theta)\ (1-\sin\theta)$

$=\left(\frac{1}{\cos\theta}+\frac{\sin\theta}{\cos\theta}\right)(1-\sin\theta)$

$=\frac{(1+\sin\theta)(1-\sin\theta)}{\cos\theta}=\frac{1-\sin^2\theta}{\cos\theta}=\frac{\cos^2\theta}{\cos\theta}=\cos\theta$

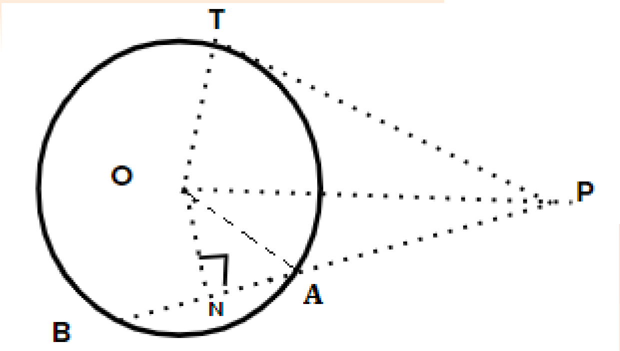
Question: If PAB is a secant to a circle with centre O intersecting the circle at A and B and PT is tangent segment, then prove that PA. PB = PT^2

Or

In a circle of radius 5 cm, AB and AC are the two chords such that AB = AC = 6 cm. Find the length of chord BC.

Solution:

(a)



Draw $ON\perp AB$ Therefore, $AN=BN$

Also PT = PA x PB ….. (1)

Consider ΔONA

$OA^2=ON^2+NA^2$ ....(2)

Similarly, in ΔPTO

$OP^2=OT^2+PT^2$

That is $PT^2=OP^2-OT^2$

$PT^2 = ON^2 + PN^2 - OA^2$ (since OA = OT (radii))

$PT^2 = ON^2 + PN^2 - ON^2 - AN^2$

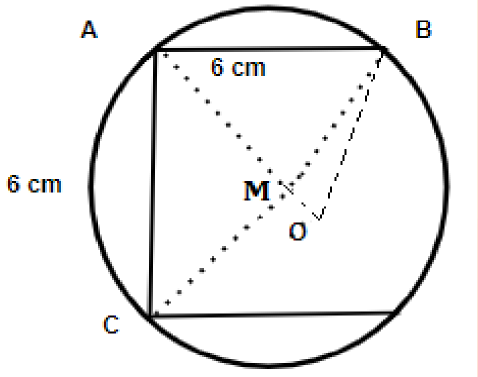
$PT^2 = PN^2 - AN^2$.........(3)

From (1) and (2) we get

$PA \times PB = PN^2 - AN^2$

$PA \times PB = (PT)^2$

(b)



$In\ \Delta ABM$

$AB^2 = BM^2 + AM^2$

$6^2 - AM^2 = BM^2$ ....(1)

In ΔBMO

$BO^2 = BM^2 + OM^2$

$5^2 - OM^2 = BM^2$.........(2)

From (1) and (2) we get,

$5^2 - OM^2 = 6^2 - AM^2$

$AM^2 = 36 - 25 + OM^2$

Since, OM = AO – AM

$AM^2 = 9 + (AO - AM)^2$

$AM^2 = 9 + (5 - AM)^2$

$AM^2 = 9 + 25 + AM^2 - 10AM$

$10AM = 36$

$AM = 3.6$ cm

$In \quad \Delta AMC$

$AC^2 = AM^2 + CM^2$

$6^2 = 3.6^2 + CM^2$

$36 - 12.96 = CM^2$

$\sqrt{23.04} = CM$

$4.8 = CM$

$\because AO \perp$ bisector of chord BC

BM = CM

BM + CM = BC

2CM = BC

2 × 4.8 = BC

BC = 9.6 cm

Question: Find the mode of the following frequency table:

Class interval	140 - 150	150 - 160	160 - 170	170 - 180	180 - 190	190 - 200
Frequency	4	6	10	12	9	3

Or

Calculate the cost of living index number for the year 1995 on the basis of year 1990 from the following data:

Item	**Quantity (in kg)**	**Cost (in Rs.) per Kg.**	
		Year1990	year 1995
A	8	30-00	45-00
B	5	28-00	36-00
C	12	6-00	11-00
D	40	9-00	15-00
E	18	10-00	12-00

(a) Modal class = 170 – 180

$$\text{Mode} = L + \frac{f_0 - f_1}{(f_0 - f_1) + (f_0 - f_2)}$$

L = 170

$f_0 = 12$

$f_1 = 10$

$f_2 = 9$

$$\text{Mode} = 170 + \frac{12 - 10}{(12 - 10) + (12 - 9)}$$

$= 170 + \frac{2}{2+3}$

$= 170 + \frac{2}{5}$

$= 170.4$